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Research Paper

MAUVE-MUSE: Survey Description and Atlas

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Abstract

This guide is for authors who are preparing papers for the *Publications of the Astronomical Society of Australia* journal using the LaTeX document preparation system and the CUP PAS style file.

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1. Introduction

2. The MAUVE sample

The MAUVE sample is drawn from the VERTICO survey (Brown et al. 2021), which in turn builds on the VIVA survey (Chung et al. 2009). The VIVA sample was primarily assembled from the 63 bright disc galaxies in the Virgo cluster presented by Koopmann et al. (2001), complemented by Hα narrow-band imaging studies by Koopmann & Kenney (2004). The original selection by Koopmann et al. (2001) aimed at achieving completeness for bright disc galaxies (S0-Scd, B_T <12 mag) in the Virgo region. However, the successive refinements leading to the VIVA and VERTICO samples have resulted in a selection function that is not clearly defined. As a consequence, these samples are not formally complete, but they are expected to capture the broad diversity of properties exhibited by satellite galaxies in Virgo.

Given the scheduling constraints associated with a large MUSE programme targeting Virgo cluster galaxies, we opted not to observe all 51 VERTICO galaxies. Instead, we restricted the sample to 40 objects while preserving its diversity in global properties. The primary selection criteria excluded galaxies undetected in CO(2-1) by VERTICO and with stellar masses (M_*, taken from Zabel et al. 2022[a].) below $10^{9.3}$ $M_\odot$, removing six galaxies in total (NGC 4299, NGC 4532, NGC 4533, NGC 4561, VCC 1581 and IC 3418).

Following a detailed assessment of the remaining sample, 5 additional galaxies were excluded as they are unlikely to have been significantly affected by the Virgo cluster environment. Specifically, NGC 4651 is a well-known example of "Umbrella" galaxy (Foster et al. 2014), which is very likely not (yet) interacting with the Virgo environment. Four further systems reside in the Virgo Southern Extension: NGC 4772 is likely a minor-merger remnant (Haynes et al. 2000; Cortese & Hughes 2009), while the remaining three galaxies show no clear evidence of environmental processing by the cluster (NGC 4536, NGC 4713 and NGC 4808 Yoon et al. 2017). As MAUVE's aim is to trace the evolutionary pathways of galaxies within the cluster environment, the exclusion of these five objects does not compromise the ability of the sample to provide a representative view of the range of environmentally driven processes affecting satellite galaxies in Virgo-like clusters.

Figure 2 shows the distribution of the MAUVE sample (circles) in the M_*–star formation rate (SFR, left) and M_*–atomic hydrogen mass (M_{HI}, middle) planes. Stellar masses and SFRs are taken from Leroy et al. (2019) and rescaled to our adopted Virgo distance of 16.5 Mpc. Atomic gas masses are obtained from Brown et al. (2021). To place MAUVE in the context of the local galaxy population and other MUSE large programmes targeting nearby galaxies, we also show galaxies from the GASP (stars; Vulcani et al. 2018) and PHANGS-MUSE (squares; Emsellem et al. 2022) samples. As a reference for the local population, we further include xGASS galaxies (Catinella et al. 2018), shown as grey dots, restricted to galaxies lying within 1σ of the star-forming main sequence. The corresponding best-fitting M_*–SFR and M_*–M_{HI} relations are shown as solid lines (Janowiecki et al. 2020). By construction, the MAUVE sample overlaps with both GASP and PHANGS-MUSE, but extends the parameter space towards significantly lower star formation rates and gas contents.

A unique aspect of the MAUVE sample is its inclusion of galaxies at different stages of environmental perturbation. While a detailed characterisation of the infall stage of each galaxy is one of the primary goals of the project, we can already exploit the past literature work to demonstrate the potential of MAUVE to construct a comprehensive infall sequence in the Virgo cluster. At the survey design stage, we divided the sample into three broad categories: galaxies that have only recently begun interacting with the Virgo environment ("pre-peak"), galaxies near the peak of environmental perturbation (e.g., in the case of ram pressure stripping a couple hundreds Myr around the peak of stripping), and galaxies that have already passed this phase ("post-peak"). Our classification is primarily based on the scheme presented by Yoon et al. (2017), which combines HI morphology and location in phase-space plane, and is further informed by the infall sequence developed by Vollmer (2009) and the work by Köppen et al. (2018), who used analytic

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[a]It recently came to our attention that the stellar masses published in this paper have not been properly rescaled by distance. Luckily, this error has not impact on our final sample selection

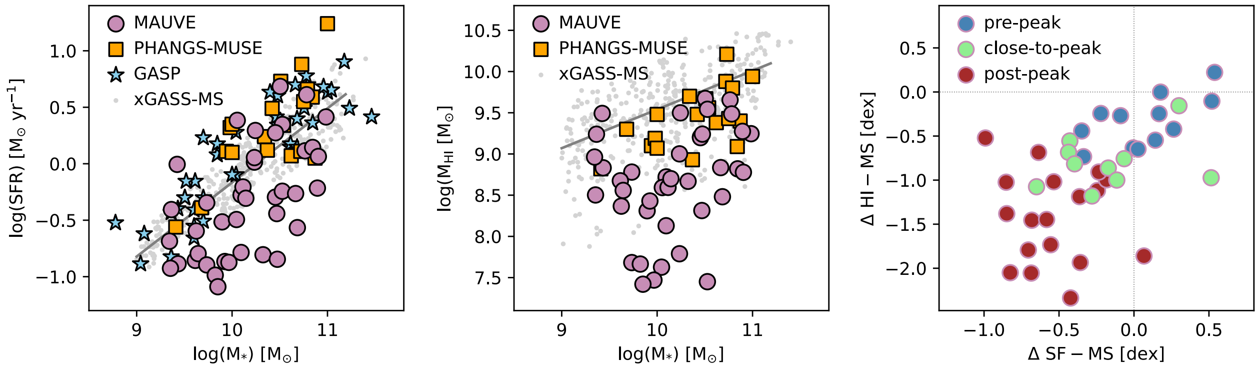

Figure 1.

models to distinguish between ongoing and past stripping events in Virgo.

Compared to Yoon et al. (2017), our main modification is to group their five classes into just three categories: pre-peak (classes 0 and I), near-peak (class II), and post-peak (classes III and IV). Only two galaxies were significantly reclassified. NGC 4192 is moved from class IV to the pre-peak category, as it is not HI-deficient and shows no clear evidence of stripping. Conversely, NGC 4698 is moved from class I to the post-peak category, as it is a peculiar system (Corsini et al. 2012), consistent with past stripping (Köppen et al. 2018), and is unlikely to represent a typical pre-peak object.

Our classification is showcased in the right panel of Fig. 2, where galaxies are colour-coded by infall stage in the ΔSF-MS–ΔHI-MS plane, which quantifies their offset from the star-forming and the gas-normal sequences. The three categories display the expected statistical trend, with decreasing SFRs and gas contents from pre-peak to near-peak to post-peak stages. The post-peak population exhibits the largest scatter, consistent with this phase representing the most diverse range of evolutionary pathways. The integrated properties and infall stage classification for the 40 galaxies in the MAUVE sample are presented in Table ??.

3. MUSE observations

VLT/MUSE observations for 37 galaxies in the MAUVE sample were obtained as part of the Large Programme (110.244E). The remaining three galaxies (NGC 4254, NGC 4321, and NGC 4535) were previously observed as part of the PHANGS-MUSE programme, and in this work we make use of the reduced datacubes publicly released by the PHANGS collaboration. Two additional galaxies (NGC 4383 and NGC 4424) have partial coverage available in the ESO archive from programmes 105.208Y (Watts et al. 2024), 097.D-0408(A) and 098.D-0115(A) (López-Cobá et al. 2020). These data were reduced as described below and combined within our MUSE mosaics.

The MAUVE-MUSE survey is designed to match the spatial coverage of the VERTICO observations, mapping the regions of the discs where molecular gas has been detected. As a result, the MUSE coverage varies substantially across the sample, ranging from small fractions of the stellar disc (∼20% of the stellar radius measured at 1M$_\odot$ pc^{-2} stellar surface density level) to nearly its full extent. Given the size of the MUSE field of view, each pointing covers a diameter of ∼4.8 kpc at the distance of Virgo, with mosaics consisting of between two and nine pointings.

MUSE was operated in Wide Field Mode with the nominal wavelength setup, providing a field of view of ∼60″×60″, a spatial sampling of 0.2″ pixel^{-1}, a spectral sampling of 1.25 Å pixel^{-1}, and a spectral resolution (FWHM) of ∼2.5 Å at 7000 Å. Observations were executed in service mode using observation blocks (OBs) of 1 hour and 13 minutes duration. Each OB consists of four on-target (O) exposures of 750 seconds each and one offset sky exposure of 249 seconds, arranged in an OOSOO sequence. Individual object exposures were rotated by 90° and combined with small dithers (<1″) to improve spatial sampling and mitigate instrumental systematics. Adjacent pointings within a given mosaic include an overlap of 5″. The archival observations adopted similar observing strategies with slightly different on-source times: 4×625 sec and 4×701 sec for the two pointings in NGC 4424; 3×750 sec and 6×750 sec for the two pointings in NGC 4383.

The MAUVE-MUSE program consisted overall of 142 OBs for a total of 173 hours. Observations started on January 17th, 2023 with the last OB observed on May 21st, 2026. To maximize schedulability, observing constraints were set to clear sky and airmass lower than 1.6. The program was originally designed to be combined with VERTICO observations with a target of a spatial resolution of 120 parsec. At the distance of Virgo, this corresponds to 1.5″. In Figure **??**, we show the distribution of the average seeing measured a Paranal during the execution of each OB. The average seeing during our observations (including archival data) was 0.73″ corresponding to a physical scale of ∼60 parsec at the distance of Virgo.

4. Data Reduction

The MAUVE-MUSE data have been reduced usinh the PYMUSEPIPE package, a Python wrapped for the MUSE data reduction system (Weilbacher et al. 2020) following the approach described in (Emsellem et al. 2022), and adopted for the PHANGS-MUSE project. This approach has become very common for the reduction of MUSE observations of nearby galaxies, with particular focus on obtaining large mosaics of extended objects. As such, in addition to the original PHANGS-MUSE paper, this approach has been extensively described in several other works (e.g., Congiu et al. 2025; Feltre et al. 2026). As such, here we highlight the main aspects that are important for the user and that may slightly differentiate our approach from previosu work.